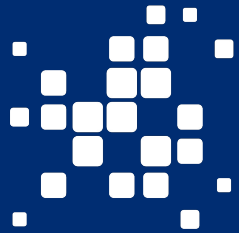
Introduction to k-mers

A brief history of k-mers in genomics
Presented by Katie



wellcome
sanger
institute

Pre-introduction introductions



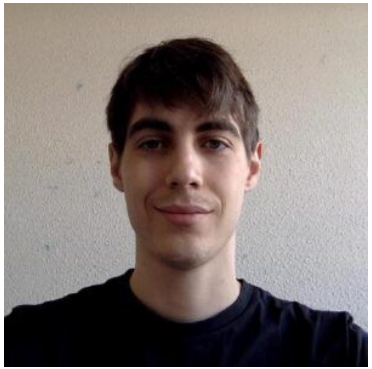
Camila Duitama Gonzalez



Richard Durbin



Kamil Jaron



Antoine Limasset



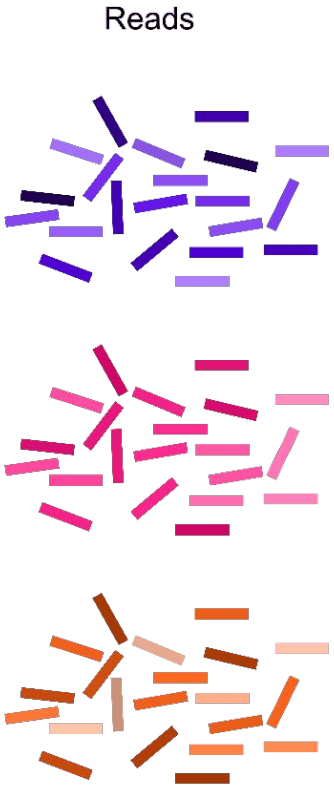
Vikram Shivkumar

wellcome sanger institute

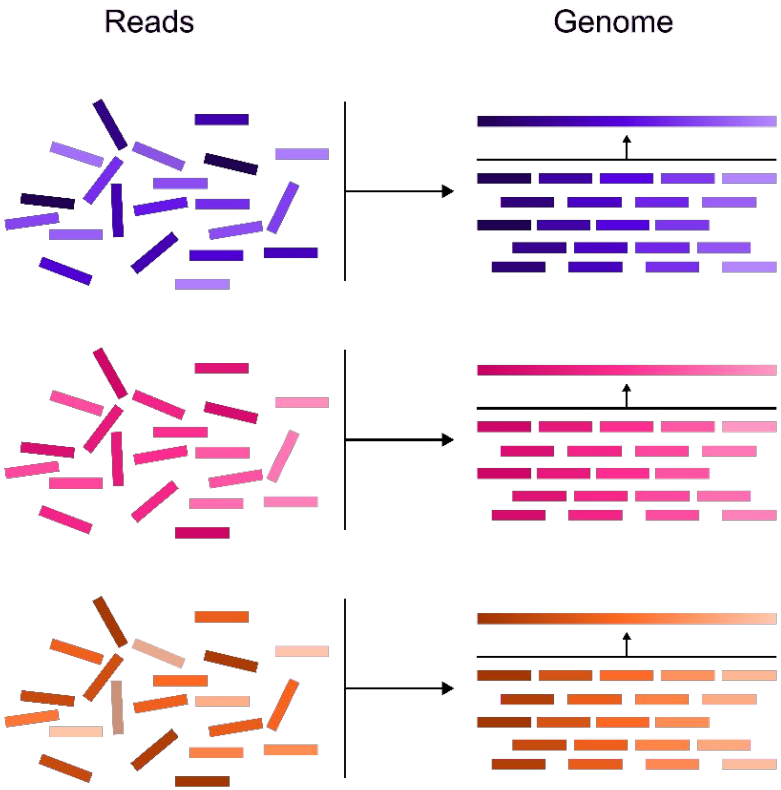


Alignment? k-mers.
Abundance estimation? k-mers
Differential analysis?
believe it or not, also k-mers (sometimes)

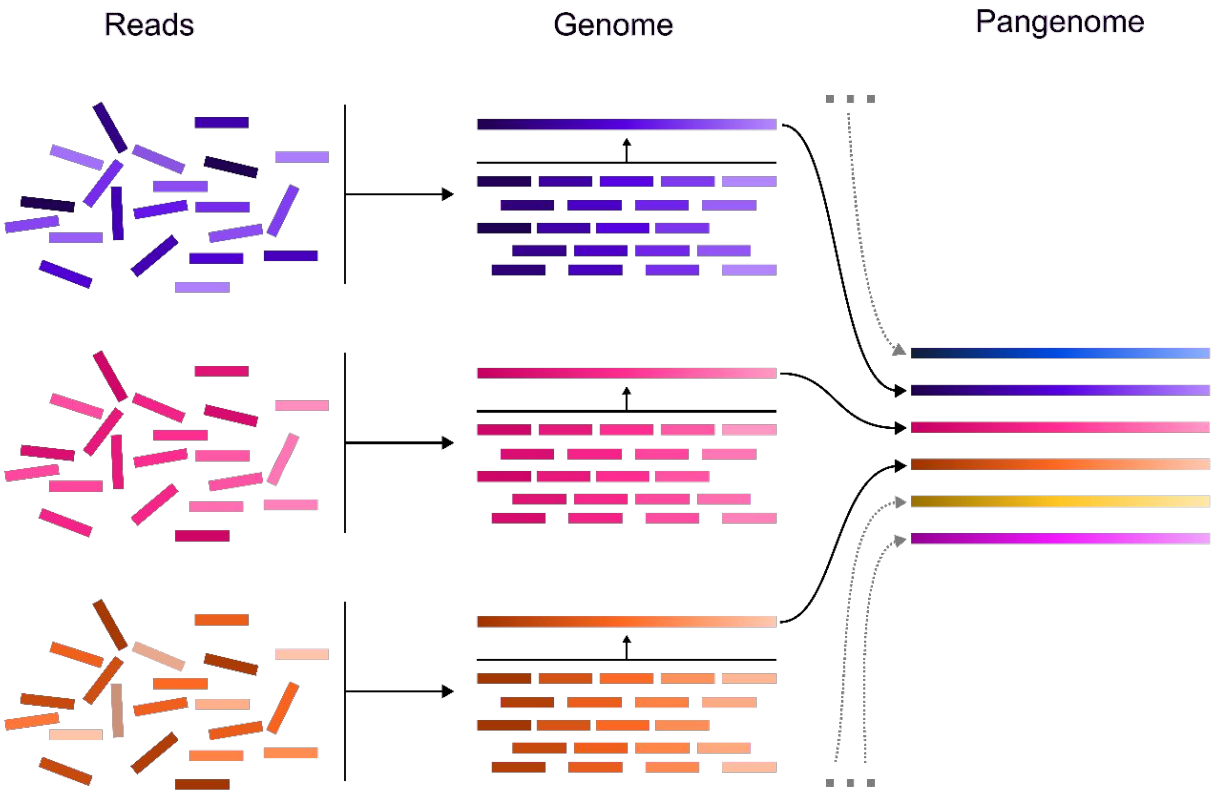
Reads → Genomes → Pangenomes



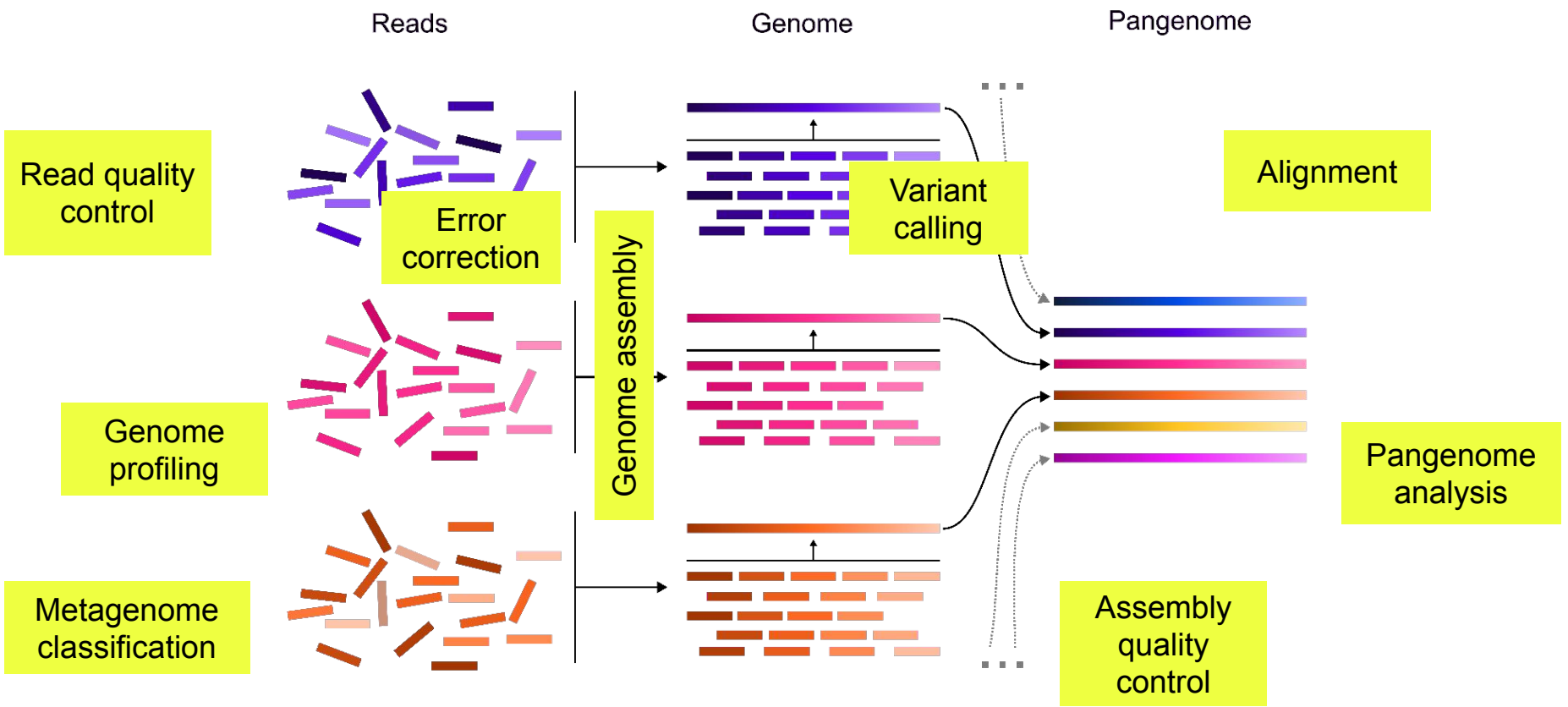
Reads → Genomes → Pangenomes



Reads → Genomes → Pangenomes



Reads → Genomes → Pangenomes



Introduction to k-mers: outline

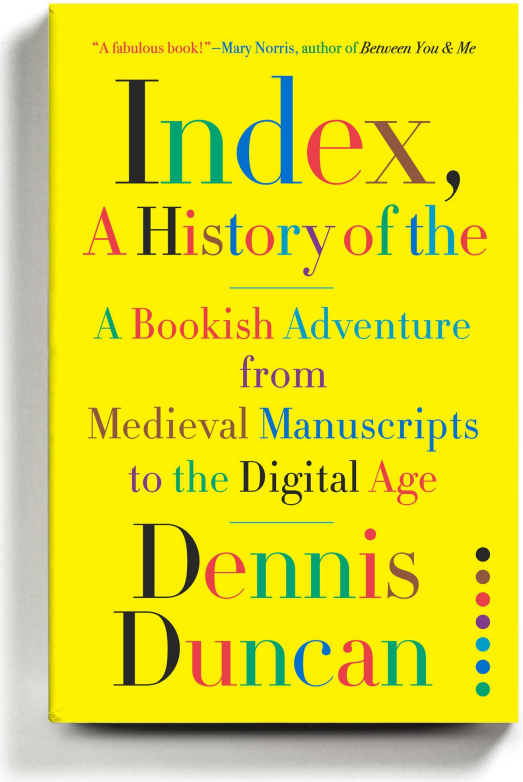
1. **What** is a k-mer
2. **Why** do we even need k-mer analysis?
3. The **building blocks** of bioinformatics
4. **Applications** for k-mers
5. **Principles** of k-mers

~~Introduction to k-mers: outline~~

Fundamentals of bioinformatics

1. **What** is a k-mer
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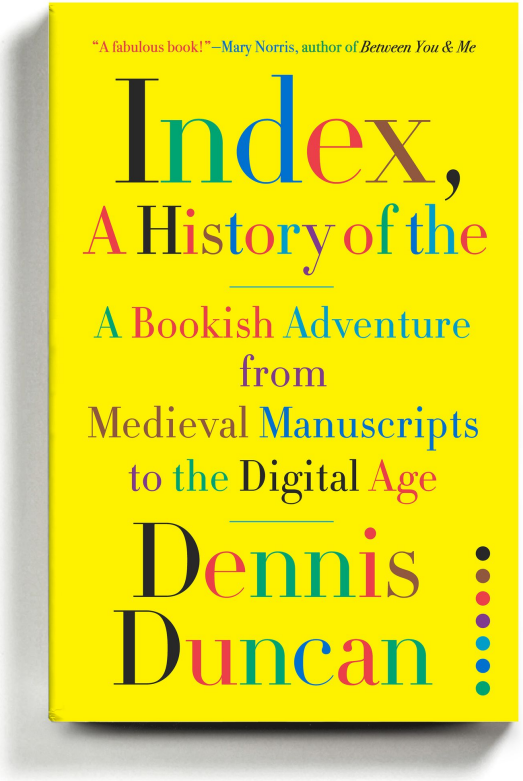
A k-mer is a substring of length k



Nelle Harper Lee (April 28, 1926 – February 19, 2016) was an American novelist whose 1960 novel *To Kill a Mockingbird* won the 1961 Pulitzer Prize and became a classic of modern American literature. She assisted her close friend Truman Capote in his research for the book *In Cold Blood* (1966).[1] Her second and final novel, *Go Set a Watchman*, was an earlier draft of *Mockingbird*, set at a later date, that was published in July 2015 as a sequel.[2][3][4]

https://en.wikipedia.org/wiki/Harper_Lee

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FASTA Formate

MN

Header → >KJ946236.1 Homo sapiens Kidd blood group protein (SLC14A1) gene, exons 4, 5 and partial cds

Sequence →
ATGGAGGACAGCCCCACTATGGTTAGAGTGGACAGCCCCACTATGGTTAGGGGTGAAAACCAGGTTTCGC
CATGTCAAGGGAGAAGGTGCTTCCCCAAAGCTCTTGGCTATGTCACCGGTGACATGAAAAAACTTGCCAA
CCAGCTTAAAGNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNN
NNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNACAAACCCGTGGTGCTCCAGTTCATTGAC
TGGATTCTCCGGGGCATATCCCAAGTGGTGTTCGTCAACGACCCCGTCAGTGGAATCCTGATTCTGGTAG
GACTTCTTGTTCAGAACCCCTGGTGGGCTCTCACTGGCTGGCTGGGAACAGTGGTCTCCACTCTGATGGC
CCTCTTGCTCAGCCAGGACAG

A k-mer is a substring of length k



Each chromosome could be represented as a very, very large string.

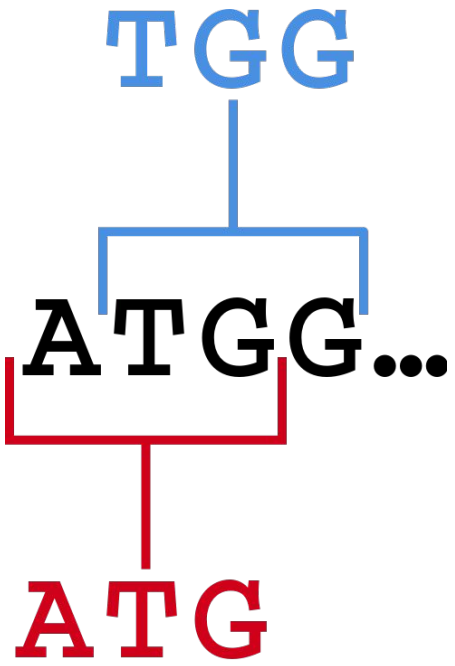
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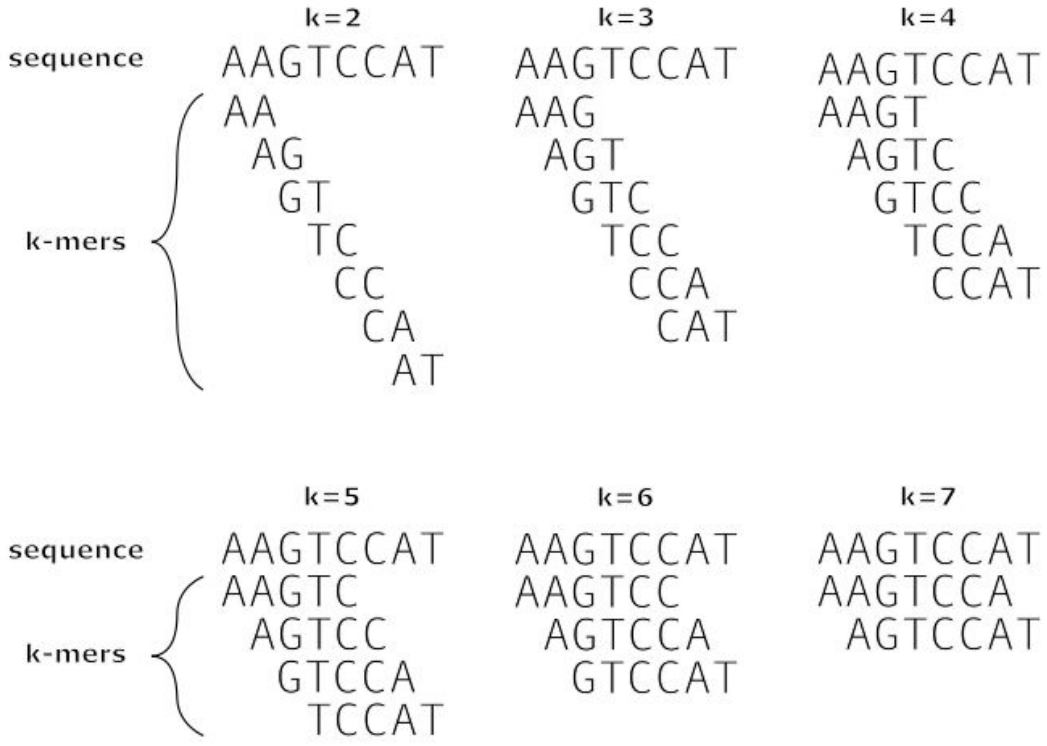
Sequence →
ATGGAGGACAGCCCCACTATGGTTAGAGTGGACAGCCCCACTATGGTTAGGGGTGAAAACCAGGTTTCGC
CATGTCAAGGGAGAAGGTGCTTCCCCAAAGCTCTTGGCTATGTCACCGGTGACATGAAAAAACTTGCCAA
CCAGCTTAAAGNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNN
NNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNACAAACCCGTGGTGCTCCAGTTCATTGAC
TGGATTCTCCGGGGCATATCCCAAGTGGTGTTCGTCAACGACCCCGTCAGTGGAATCCTGATTCTGGTAG
GACTTCTTGTTCAGAACCCCTGGTGGGCTCTCACTGGCTGGCTGGGAACAGTGGTCTCCACTCTGATGGC
CCTCTTGCTCAGCCAGGACAG

A k-mer is a substring of length k



Sequence ATGG has two 3-mers: ATG and TGG.

A k-mer is a substring of length k

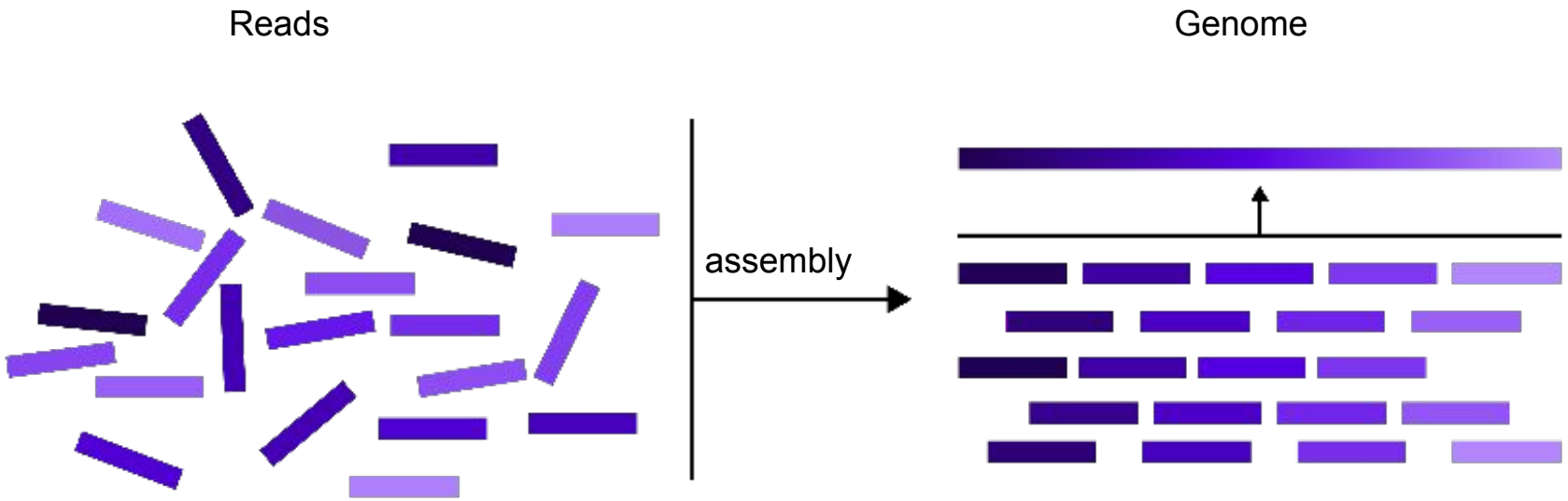


https://github.com/KamilSJaron/k-mer-approaches-for-biodiversity-genomics/wiki/Concept-of-k-mers

Introduction to k-mers: outline

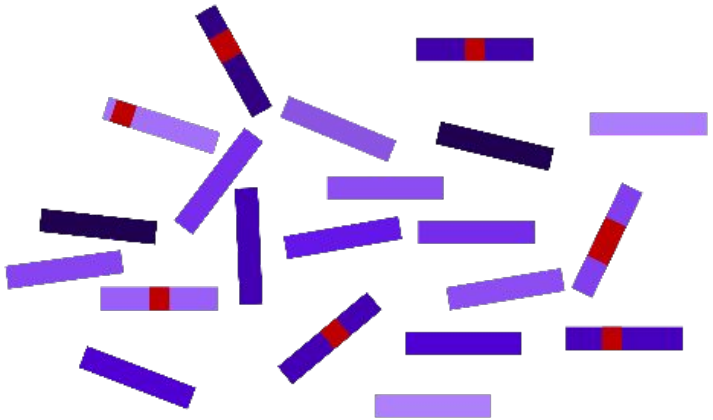
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Genome assembly example: Why breaking reads into smaller pieces is actually a good thing



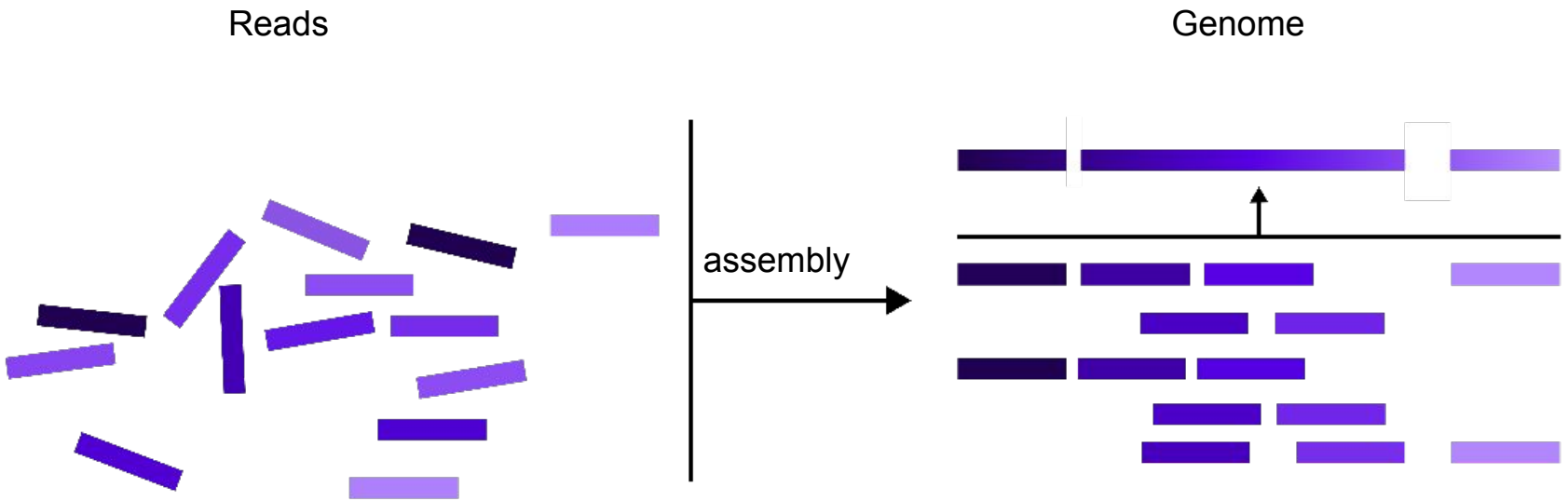
Genome assembly example: Why breaking reads into smaller pieces is actually a good thing

Reads



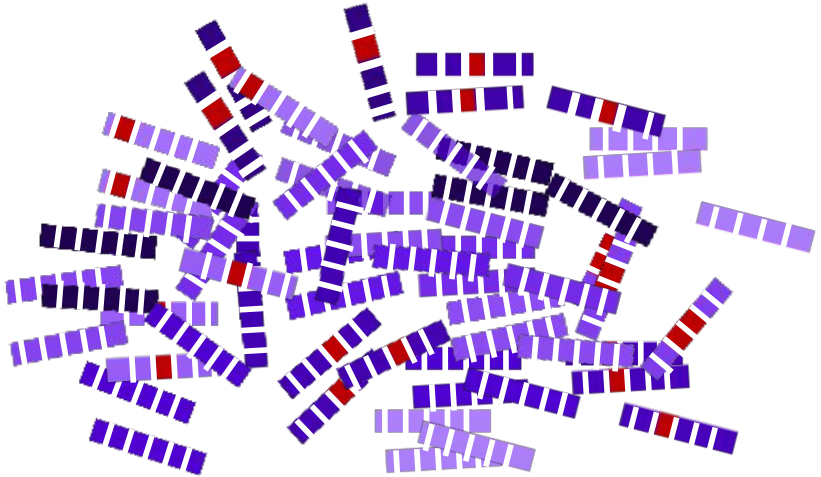
✖ Sequencing error

Genome assembly example: Why breaking reads into smaller pieces is actually a good thing



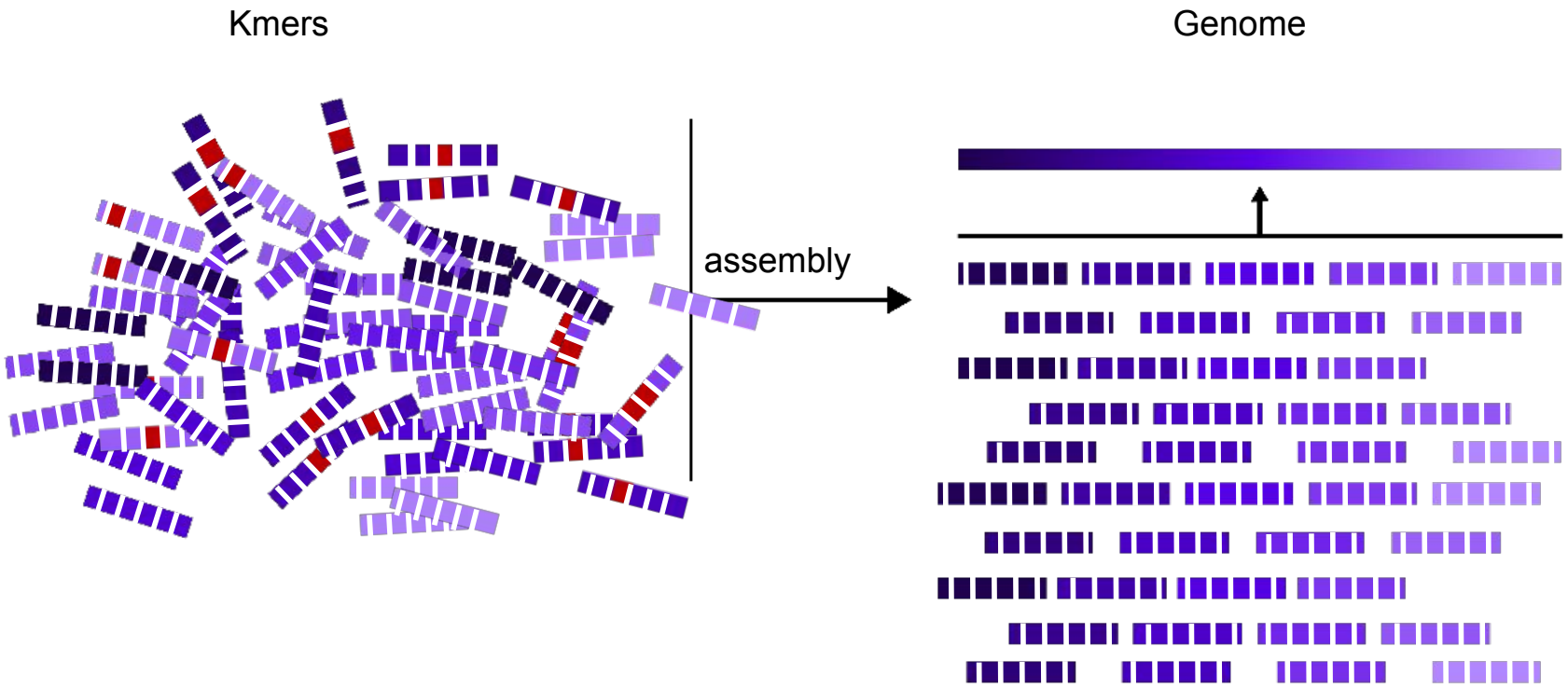
Genome assembly example: Why breaking reads into smaller pieces is actually a good thing

Kmers



Genome

Genome assembly example: Why breaking reads into smaller pieces is actually a good thing



Why breaking reads into smaller pieces is actually a good thing

- K-mers allow efficient indexing and searching. Think of an index in a book

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- Simplifies complex regions

Why breaking reads into smaller pieces is actually a good thing

- K-mers allow efficient indexing and searching. Think of an index in a book
- Simplifies complex regions
- Converts long, complicated, error prone sequence into digestible and standardized units that can be efficiently compared.

Introduction to k-mers: outline

1. **What** is a k-mer
2. **Why** do we even need k-mers?
3. **The building blocks of bioinformatics**
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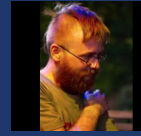


Alignment? k-mers.
Abundance estimation? k-mers
Differential analysis?
believe it or not, also k-mers (sometimes)

History of k-mers

- **N-gram** (Shannon, 1948)
- **k-tuple** (Drmanac et al. 1991; Idury and Waterman 1995)
- **ktup** (Lipman and Pearson 1985)
- **L-tuple** (Idury and Waterman 1995)
- **ℓ-tuple** (Li and Waterman 2003)
- **k-word** (Lippert et al. 2002; Li and Waterman 2003)
- **word** (Reinert et al. 2000)
- **11-mers** (Drmanac et al. 1989)
- **w-mers** (BLAST, 1990)
- **k-mer** (MUMmer, 1999)

History of k-mers

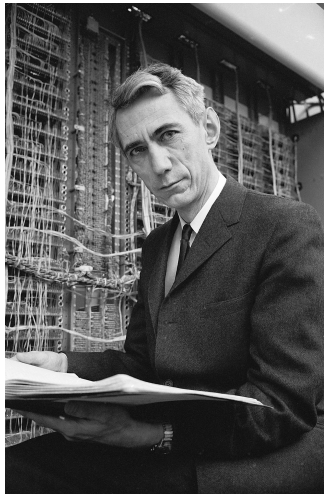


N.G. de Bruijn (1946)
de Bruijn sequences
Construct a short text that contains all k-mers



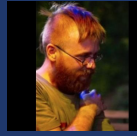
B and C Shannon
C. Shannon (1948) Information theory
Predict what character is likely to follow a given k-mer





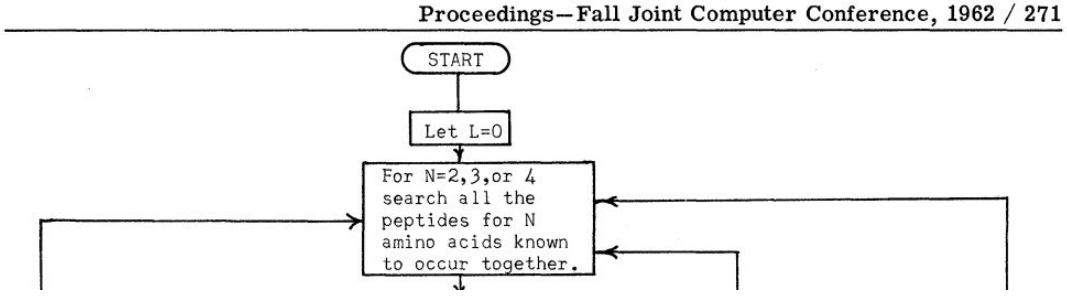
“Some of his early papers and even later papers are in my handwriting … and not in his, which confused people at first.” - Betty Shannon

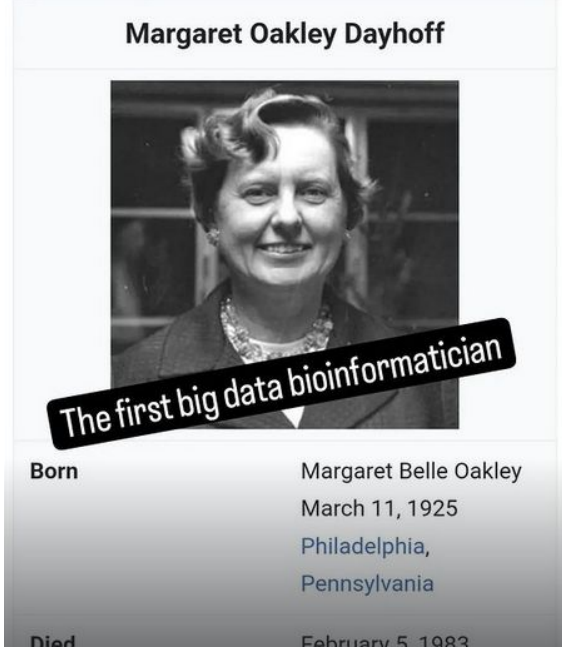
History of k-mers: K-mers introduced to bioinformatics



Margaret Dayhoff:

- k-mer search in “COMPROTEIN: A Computer Program to Aid Primary Protein Structure Determination.” 1962

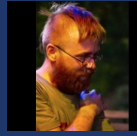




Introduction to k-mers: outline

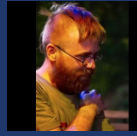
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Early applications: similarity searches



- **Smith-Waterman** (1981): local alignment (no k-mers)
- **FASTA** (1985): heuristic for faster search
  - used short exact matches as seeds — conceptually close to k-mers.
- **GenBank** launched (1979): early sequence databases
- Conceptual precursors to k-mers : “motifs” & “short matches”

FASTA and the use of 2-mers as seeds



- **FASTA** (Lipman, Pearson, 1985)
  - k=2 for proteins
  - k=4,6 for nucleotides

1) Identifies regions with high density of k-mer matches
2) Run Smith-Waterman alignment on them

The k-mer matches do not serve as anchors for the alignment.
(They serve to “light up” the right region(s))

-tuples or words? Make up your mind!



"ordered list" or "finite sequence"



I love chaos!

Michael Waterman

"k-tuple"
"L-tuple"
"ktup"

Idury RM, Waterman MS. A new algorithm for DNA sequence assembly. J Comput Biol. 1995;2: 291–306.

Lipman DJ, Pearson WR. Rapid and sensitive protein similarity searches. Science. 1985;227: 1435–1441.

"ℓ-tuple"

Li X, Waterman MS. Estimating the Repeat Structure and Length of DNA Sequences Using ℓ-Tuples. Genome Res. 2003;13: 1916–1922.

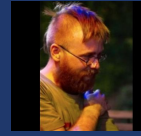
"k-word"

Lippert RA, Huang H, Waterman MS. Distributional regimes for the number of k-word matches between two random sequences. Proc Natl Acad Sci U S A. 2002;99: 13980–13989.

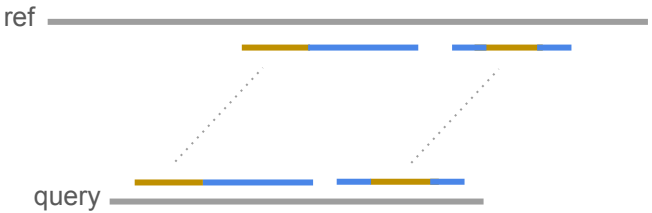
"word"

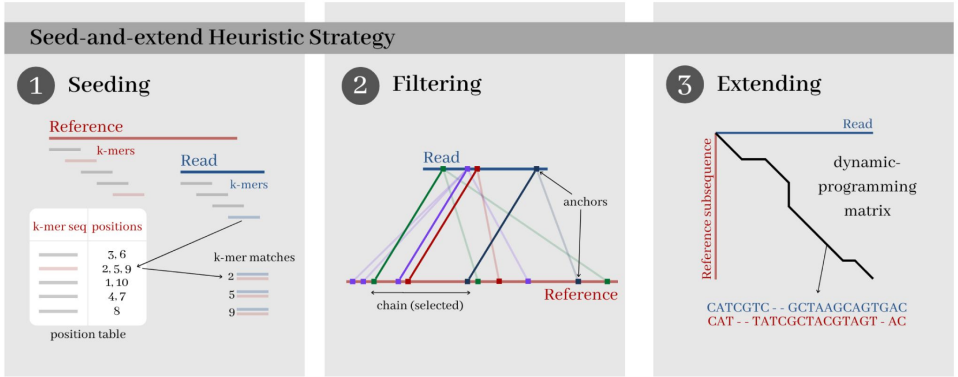
Reinert G, Schbath S, Waterman MS. Probabilistic and statistical properties of words: an overview. J Comput Biol. 2000;7: 1–46.

K-mers for alignment seeds



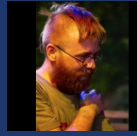
1. Find seeds
2. Filter (or chain) seeds
3. Extend alignment to the left/right of seeds using exact algorithm (Smith-Waterman)





Teng et al, 2023

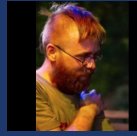
BLAST: longer seeds



- **BLAST** (Altschul, Gish, Miller, Myers, Lipman, 1990)
- Improves upon FASTA
- Uses k-mers as seeds
    - k=11,12 for blastn
- Then runs Smith-Waterman around the seeds

Compare to FASTA which has k=6.

-mer enters our vocabulary



"Mer" comes from "meros" which means "a share"

1968: "Unique Sequences in Tobacco Mosaic Virus"

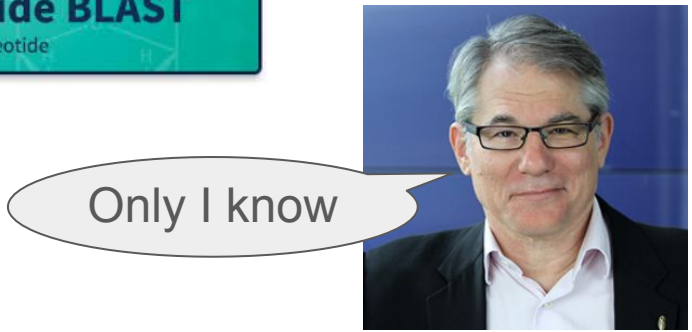
oligomers
"Ψ-mer" and "Ω-mer"

Is Stanley Mandeles father of *k*-mers?!

1990: BLAST uses "w-mers"

- Is the "w" for "words" here?

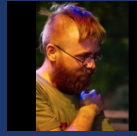




1999: MUMmer

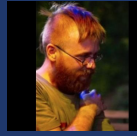
2010: Jellyfish; *k*-mers reach fixation

K-mers in alignment



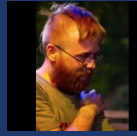
- Reads-vs-genome
  - **Bowtie1** (Langmead et al, 2009)
    - k=28 as seed, with 2 mismatches
- Genome-vs-genome
  - **LASTZ** (Bob Harris)
    - Uses spaced seeds, k=19 with 8 mismatches
  - **YASS** (Noe, Kucherov, 2005)
    - Uses spaced seeds too
- Protein-vs-protein
  - **DIAMOND** (Bunchfink, Huson, 2015)
    - Uses a set of spaced seeds

Methods with super short k-mers (k=4, 5, 6…)



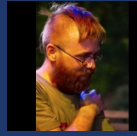
- Metagenomic binning tools
  - They look at nucleotide composition, to group contigs into genomes
  - **TETRA** (Teeling et al, 2004): **k=4**
  - Generalizes over GC content (k=1)
  - Many others too: MetaBAT2, MaxBin 2
- Metagenomics taxonomic classification
  - **QIIME2**'s q2-feature-classifier (Bukolich et al, 2018): **k>=4**
  - Naive Bayes classifier on k-mer counts
- Virus binning
  - **VirFinder** (Ren et al, 2017): **k=4**
- Genome masking
  - Komplexity (Clark 2019): **k=4**
  - Removes low-complexity regions

K-mers for analyzing genome structure



- Genome size and repeats (Li and Waterman, 2003)
  - Models k-mer histograms (without calling them that) as a mixture of Poisson distributions
  - Reconstructs repeat consensus sequences(!)
- Many follow-up works in modeling genome histograms
  - (Chor et al 2009, Liu et al 2013, Chikhi and Medvedev 2014, SPAdes assembler, Sun et al 2018, etc..)
- **CRT** (Bland et al, 2007)
  - Finds location of CRISPRs using exact k-mer matching
- **Tallymer** (Kurtz et al, 2009)
  - Find Transposible Elements in plant genomes using k-mers

K-mers in vintage genome assemblers



- **Sequencing by Hybridization** (1987)
  - Basically a microarray of k-mers
  - k=8
- String graphs and **de Bruijn graphs**, both introduced at the same conference (DIMACS 1994)

  A History of DNA Sequence Assembly, G. Myers, 2016
  - **Idury-Waterman** (1995 journal version)
    - de Bruijn graph theory for assembly (without naming it)
- **EULER** (Pevzner, Tang, Waterman 2001)
  - first assembly tool using a dBG
- Many many more tools: EULER-SR, Velvet, SOAPdenovo, SPAdes,

IDURY AND WATERMAN

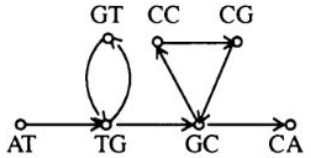
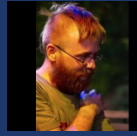


FIG. 2. Graph G for ATGTGCCGCA.

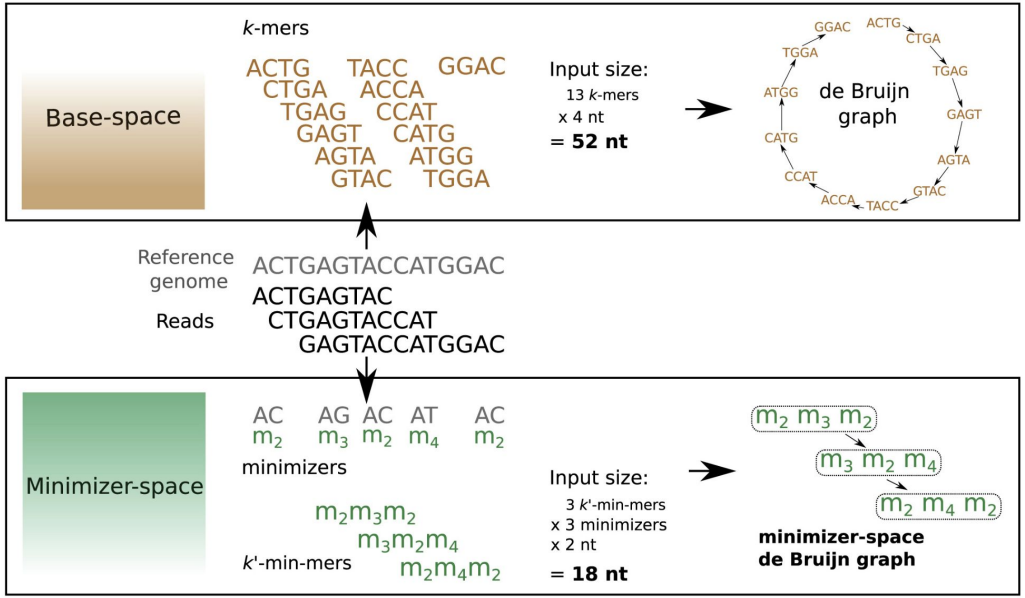
De Bruijn graphs, **unitigs**, **contigs**: will be covered later in the course (Logan lecture)

Choice of **best k:** also covered later (“In-depth understanding” lecture)

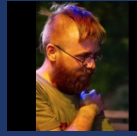
K-mers in recent genome assemblers



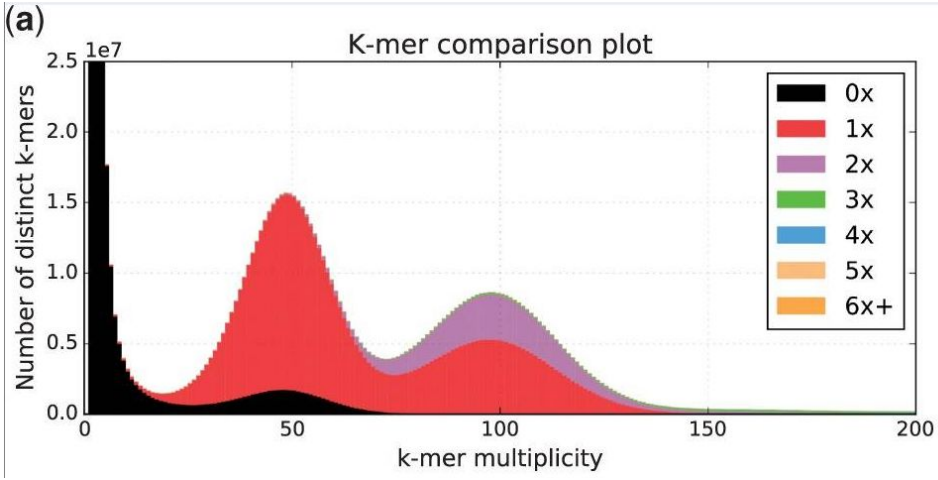
- **Verkko, LJA**
  - Apply long k-mers to long read assembly
- **rust-mdBG**
  - Uses k-mers over a different alphabet (minimizers)
- **metaMDBG**
  - State-of-the-art metagenomics assembly based on mdBG

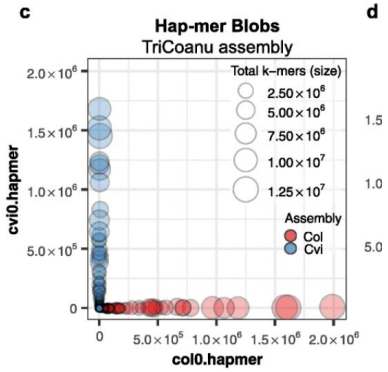


K-mers for genome quality assessment

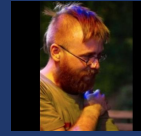


- **KAT** (Mapleson et al, 2016)
  - Compares k-mers in short reads vs k-mers in assembly
  - Or GC-content vs k-mer frequeny
- **QUAST-LG** (Mikheenko et al, 2018)
  - Uses k-mers to estimate completeness
- **Mercury** (Rhie et al, 2020)
  - Improves upon KAT
  - Extended to phased diploid genomes
  - Generates a consensus quality QV value



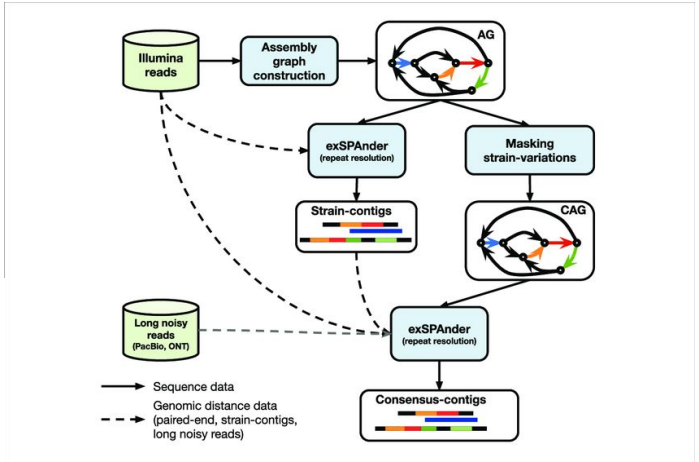


K-mers in metagenomics



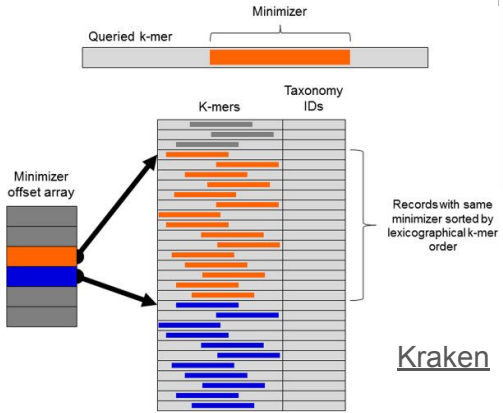
Metagenomic short read assembly: dominated by k-mers

- **metaSPAdes, MEGAHIT, minia3**



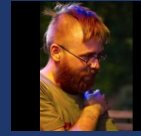
Taxonomic classification: same

- **Kraken, Centrifuge**



Kraken

K-mers in read error correction (for genome assembly)

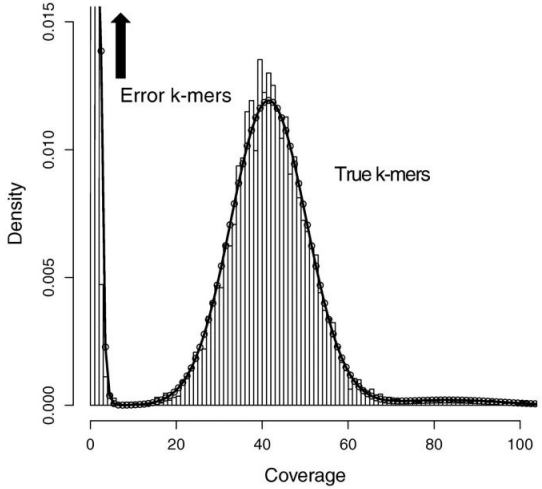


Sanger reads error correction

- **EULER**'s error correction module (Pevzner, 2001)
  - Mutates reads so that all k-mers are solid

Short reads:

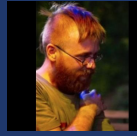
- **Quake** (Kelley, Schatz, Salzberg, 2010)
  - First use of k-mer histogram to determine error threshold
  - Frames EULER's method as a Maximum Likelihood
- **BFC**, **Lighter**, Musket, Bloocoo, (Bayes)HAMMER, ..
- Nowadays: none used anymore



Long reads:

- **LoRDEC, LoRMA**
  - de Bruijn graph to correct long reads using short reads (or long reads)

K-mers for RNA-seq analysis



Gene quantification:

- **Salmon** (Patro et al, 2017)
- **Kallisto** (Bray et al, 2016)

RNAseq specific mapping:

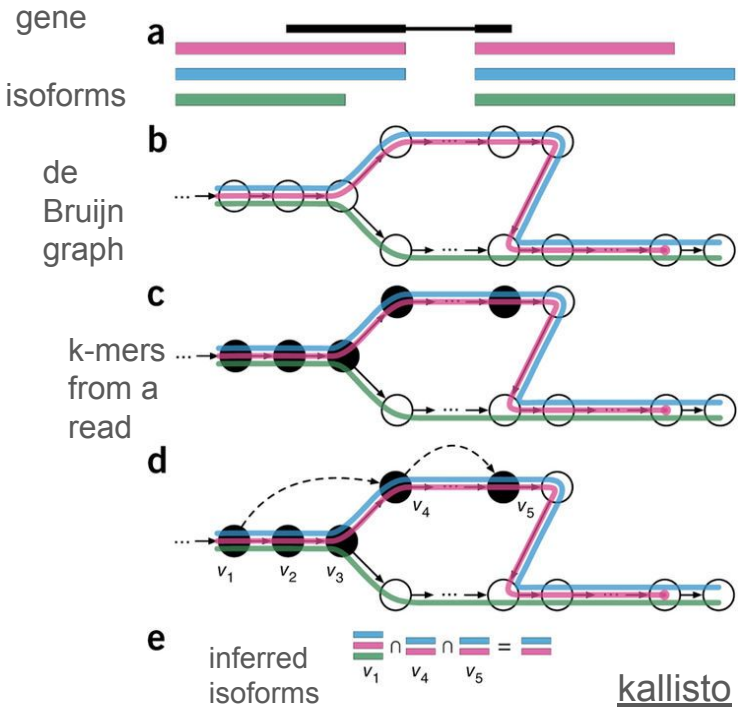
- **STAR** (Dobin et al, 2012)

RNAseq assembler:

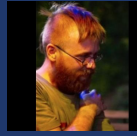
- **Trinity** (Grabherr et al, 2011)

Reference-free diff analysis:

- **Transipedia.org** (Bessière et al, 2011)



K-mers in ML



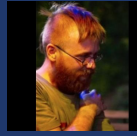
Understanding Transformers via N-gram Statistics

Timothy Nguyen
Google DeepMind
timothycnguyen@google.com

How does a transformer-based large language model (LLM) make use of its context when predicting the next token?

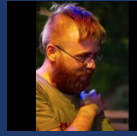
Recall Shannon's theory: probably of next gram following a N-gram

K-mers and sketching



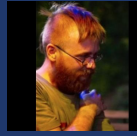
- **D_2** (Torney et al, 1990)
  - number of shared k-mers as a proxy of sequence similarity
- **MinHash** (Broder, 1997)
  - Measuring similarity between two documents by random sampling of words
- **Winnowing/minimizers** (Roberts, 2004)
  - Sampled k-mers
- **Bloom filters** (Bloom, 1970) applied to k-mers (2004)
  - First applied to accelerate BLAST
  - Later to k-mer counting, error correction, assembly, etc..
- **Mash/sourmash/Dashing**
  - Modern k-mer sketching for sequence similarity estimation

K-mers in sequence database search



- **BLAST** (1990)
  - Clustered GenBank, assembled sequences, 200 GB
- **BigSI/COBS** (Bradley et al, 2019)
  - Isolate genomes, 170 TB (.3% of SRA)
- **MetaGraph** (Karasikov et al, 2024 preprint)
  - Animals, Plants, Bacteria, cancer data, 3.3 Pbp (8.9% of SRA)
- And many more

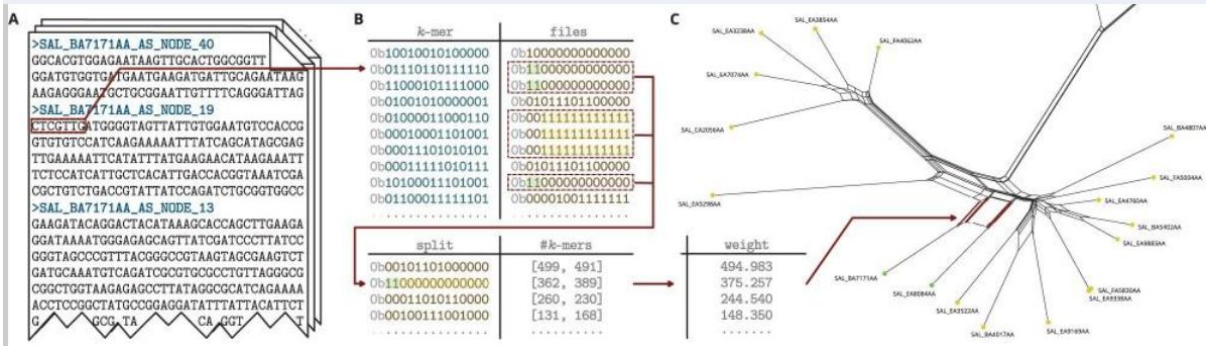
K-mers in phylogeny



- Construct phylogenetic trees without multiple sequence alignment
  - “Is Multiple-Sequence Alignment Required for Accurate Inference of Phylogeny” (Hölh, Ragan, 2007)
  - “*we encode K-mers as character states and estimate posterior probabilities of bipartitions using Mr-bayes*”

Recent works close to pangenomics:

- **SANS, SANS serif** (Rempel, Wittler, 2021), **KINN** (Tang et al, 2023),



Introduction to k-mers: outline

1. **What** is a k-mer
2. **Why** do we even need k-mers?
3. The **building blocks** of bioinformatics
4. **Applications** for k-mers
5. **Principles** of k-mers

K-mer counting



Sequence of length L (L=21)

A C A C T C A T G C T A G C T T C A A T A

K-mer counting



Sequence of length L (L=21)

A C A C T C A T G C T A G C T T C A A T A

k=4

A C A C
C A C T
A C T C
C T C A
T C A T
C A T G
A T G C
T G C T
G C T A
C T A G
T A G C
A G C T
G C T T
C T T C
T T C A
T C A A
C A A T
A A T A

K-mer counting



Sequence of length L (L=21)

A C A C T C A T G C T A G C T T C A A T A

k=4

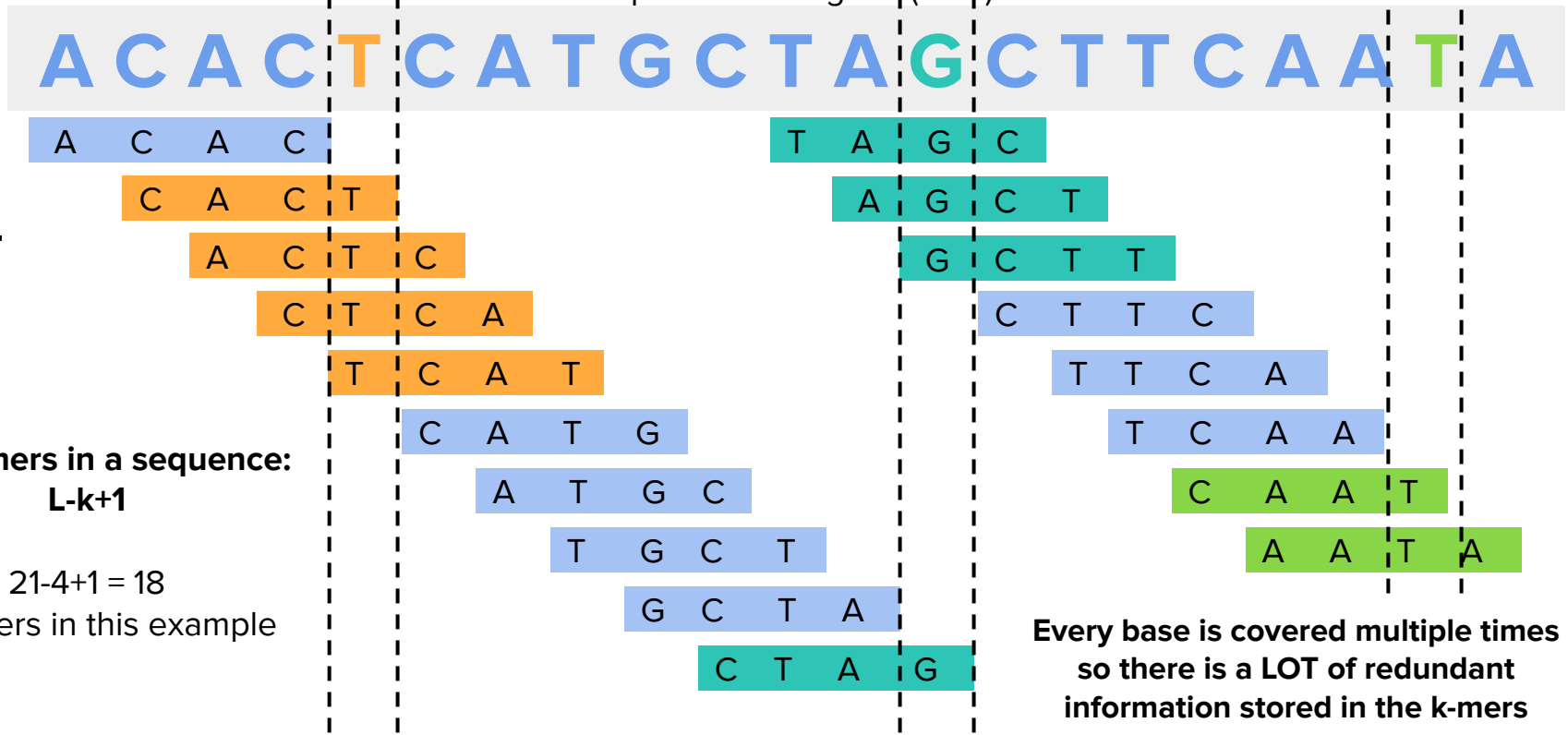
A C A C
C A C T
A C T C
C T C A
T C A T
C A T G
A T G C
T G C T
G C T A
C T A G
T A G C
A G C T
G C T T
C T T C
T T C A
T C A A
C A A T
A A T A

Every base is covered multiple times so there is a LOT of redundant information stored in the k-mers

K-mer counting



Sequence of length L (L=21)

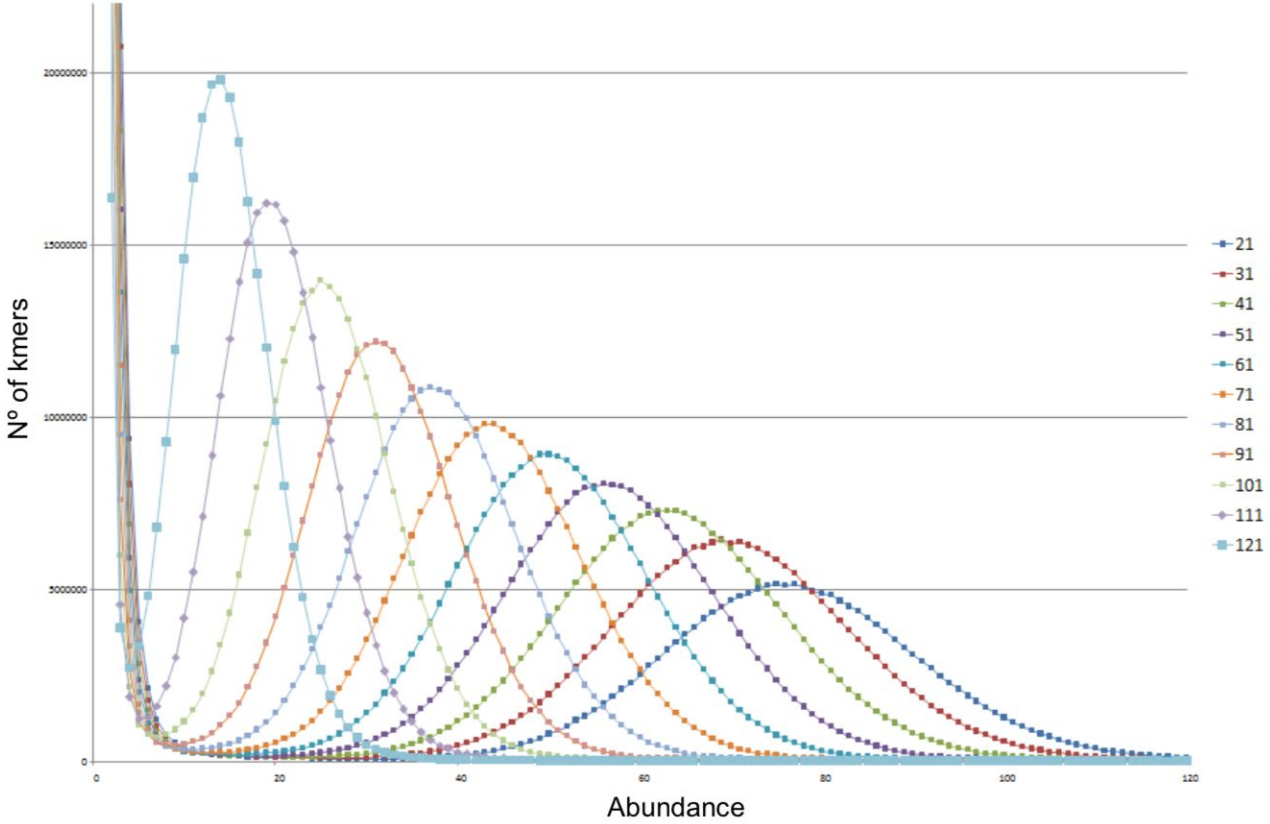


of k-mers in a sequence:
L-k+1

21-4+1 = 18
18 4-mers in this example

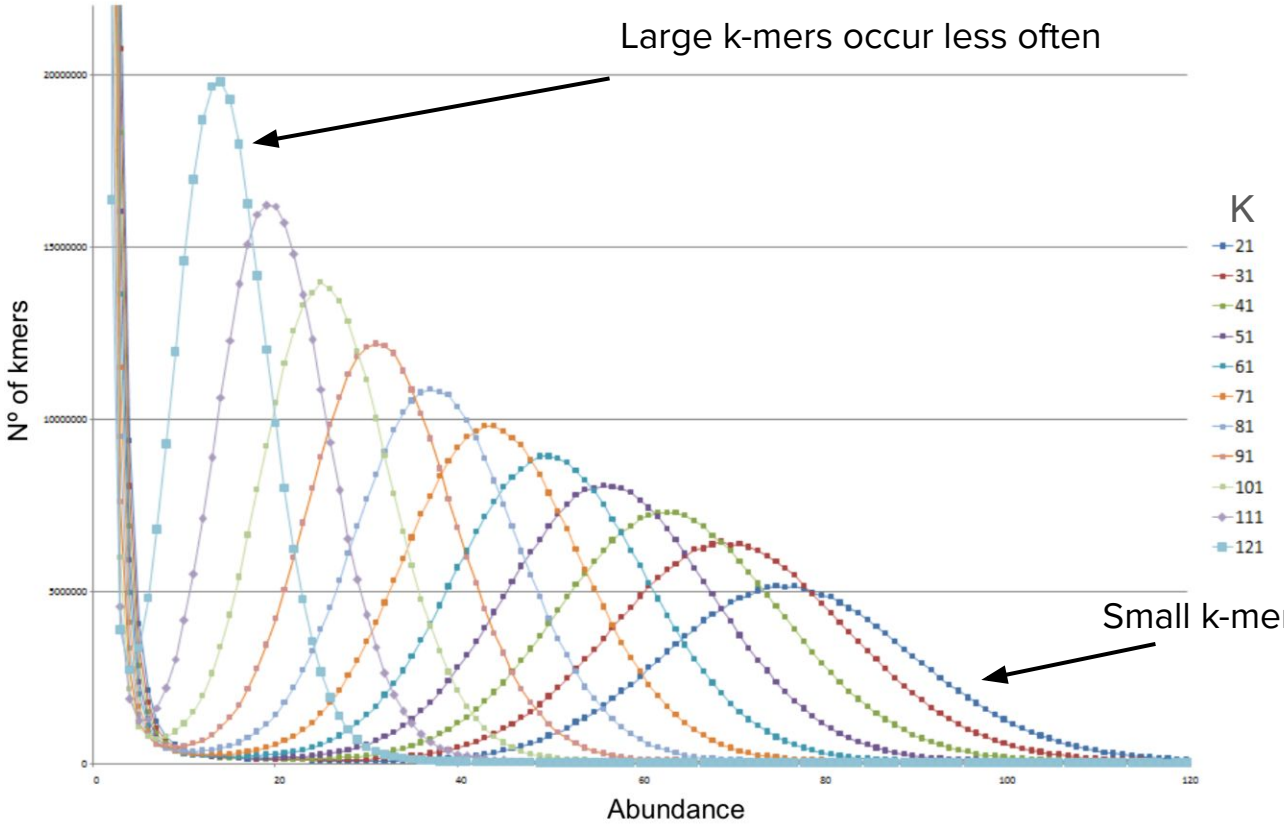
Every base is covered multiple times so there is a LOT of redundant information stored in the k-mers

K-mer counting





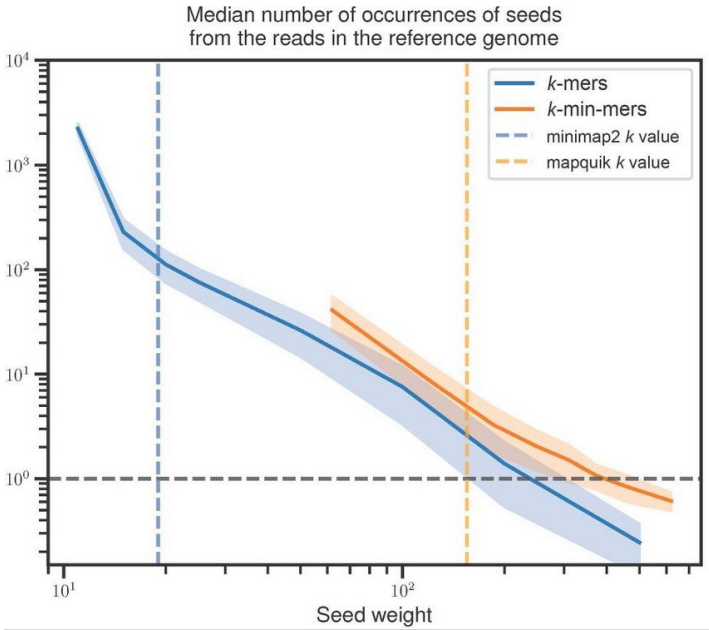
K-mer counting





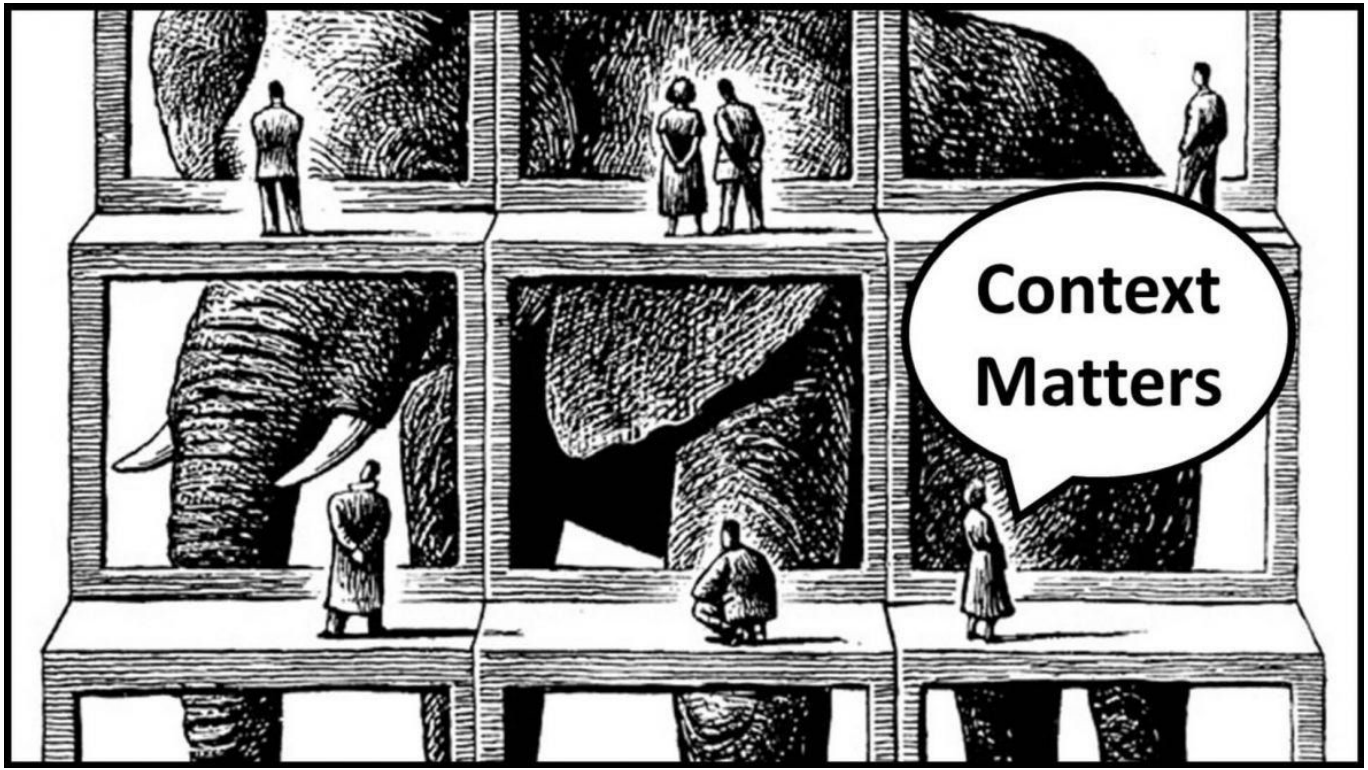
Balancing k-mer size for alignment speed vs sensitivity

- **Long** k-mer seeds:
  - high specificity (= few candidate alignment locations)
  - low sensitivity (= less likely to find the right location)
- **Short** k-mer seeds:
  - low specificity (= many potentially wrong options)
  - high sensitivity (= more likely to find the right location)



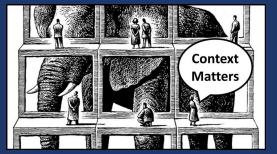
Ekim et al, 2023 (mapquik, human genome)

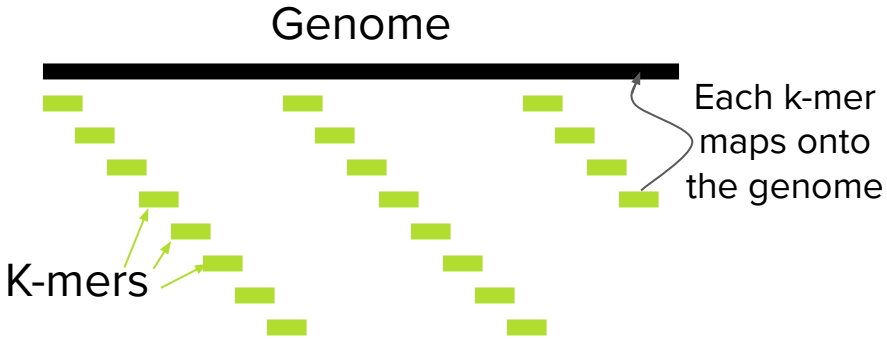
K-mer contexts



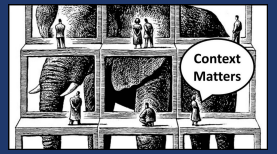
[Photo: inspectorinsight.com]

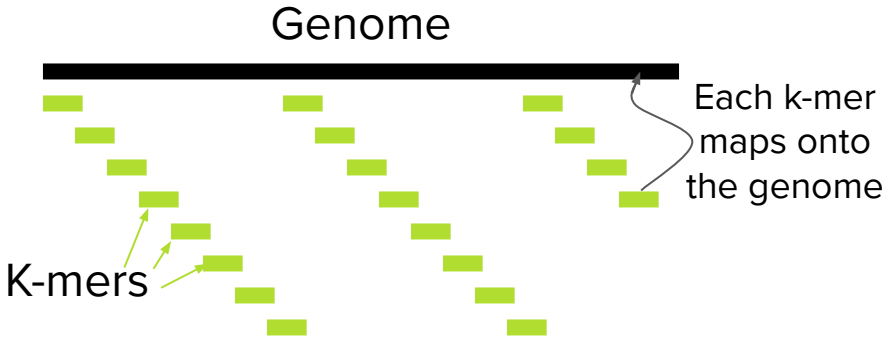
Genome k-mers vs read k-mers

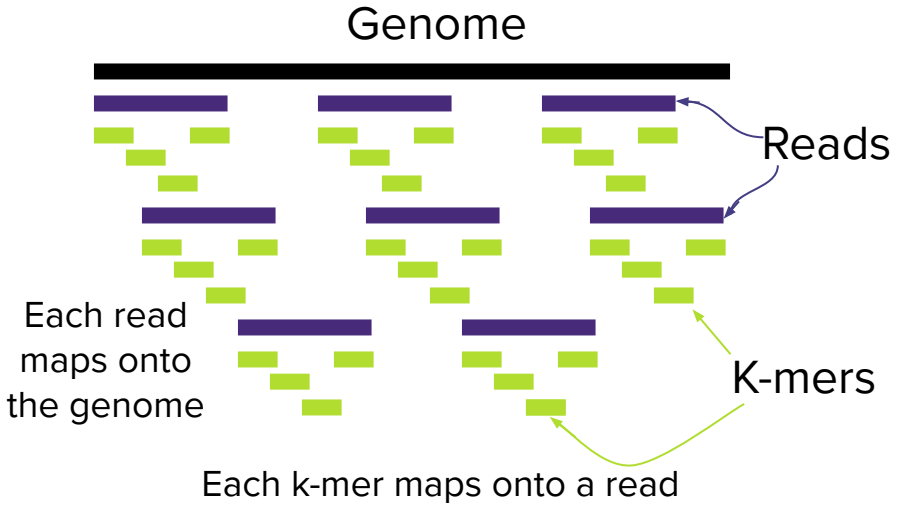




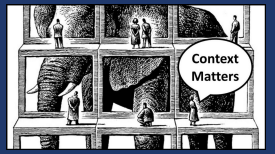
Genome k-mers vs read k-mers

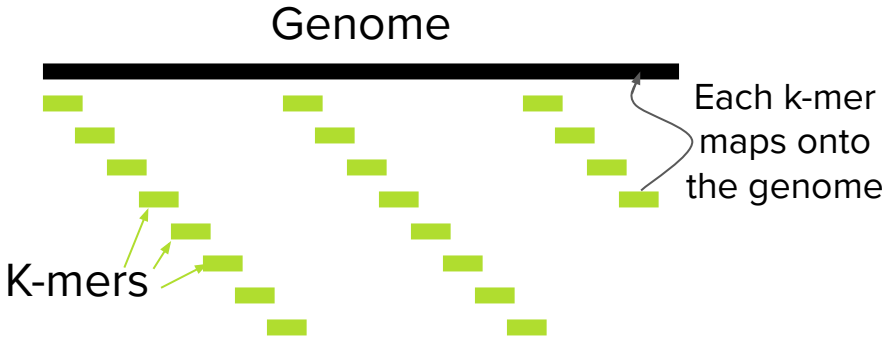


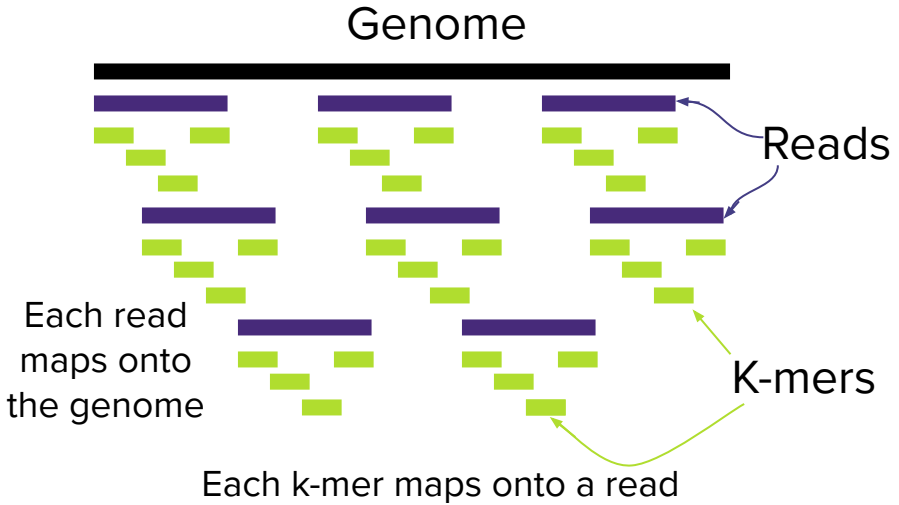




Genome k-mers vs read k-mers

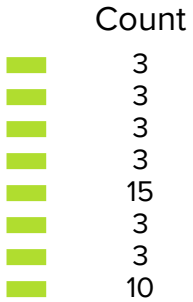








Higher counts reflect **repetitive sequence** (occurs multiple times throughout the genome)



Higher counts reflect **higher coverage** of that region and/or more **repetitive sequence**

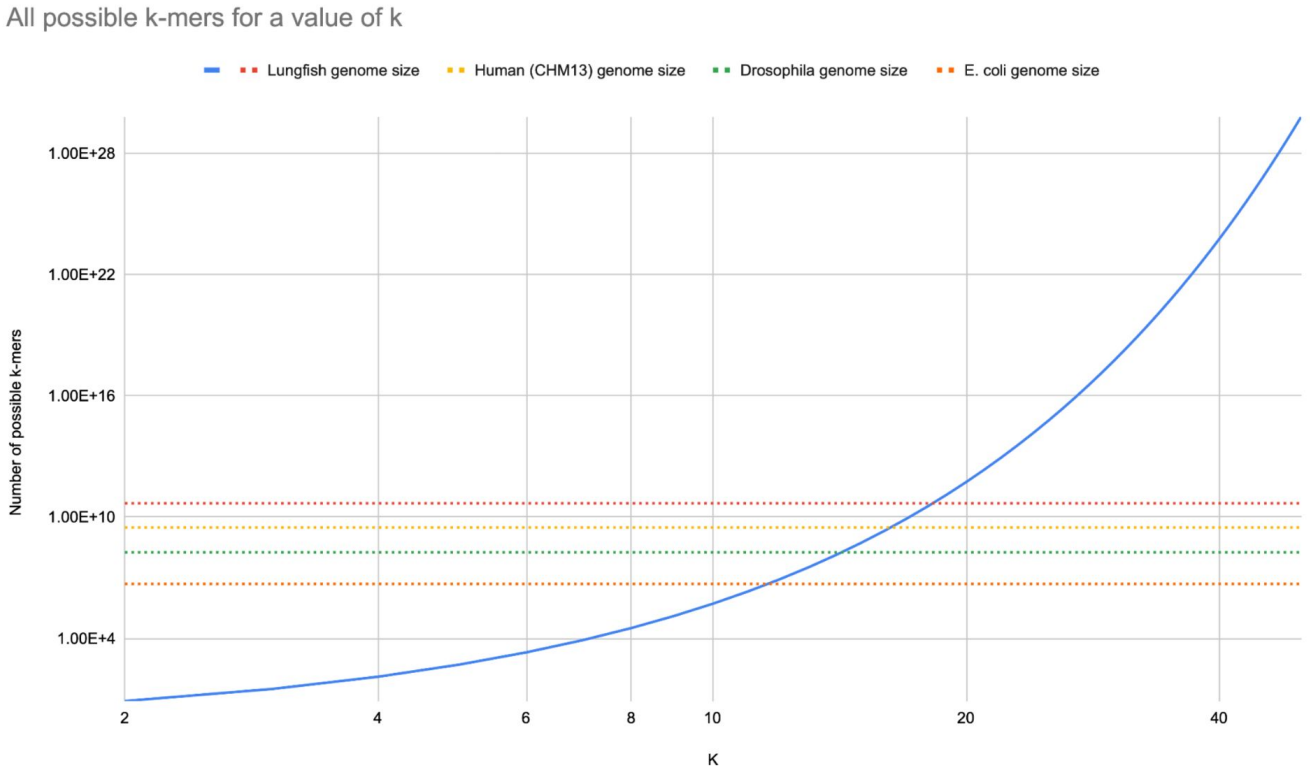
Describing the k-mer space



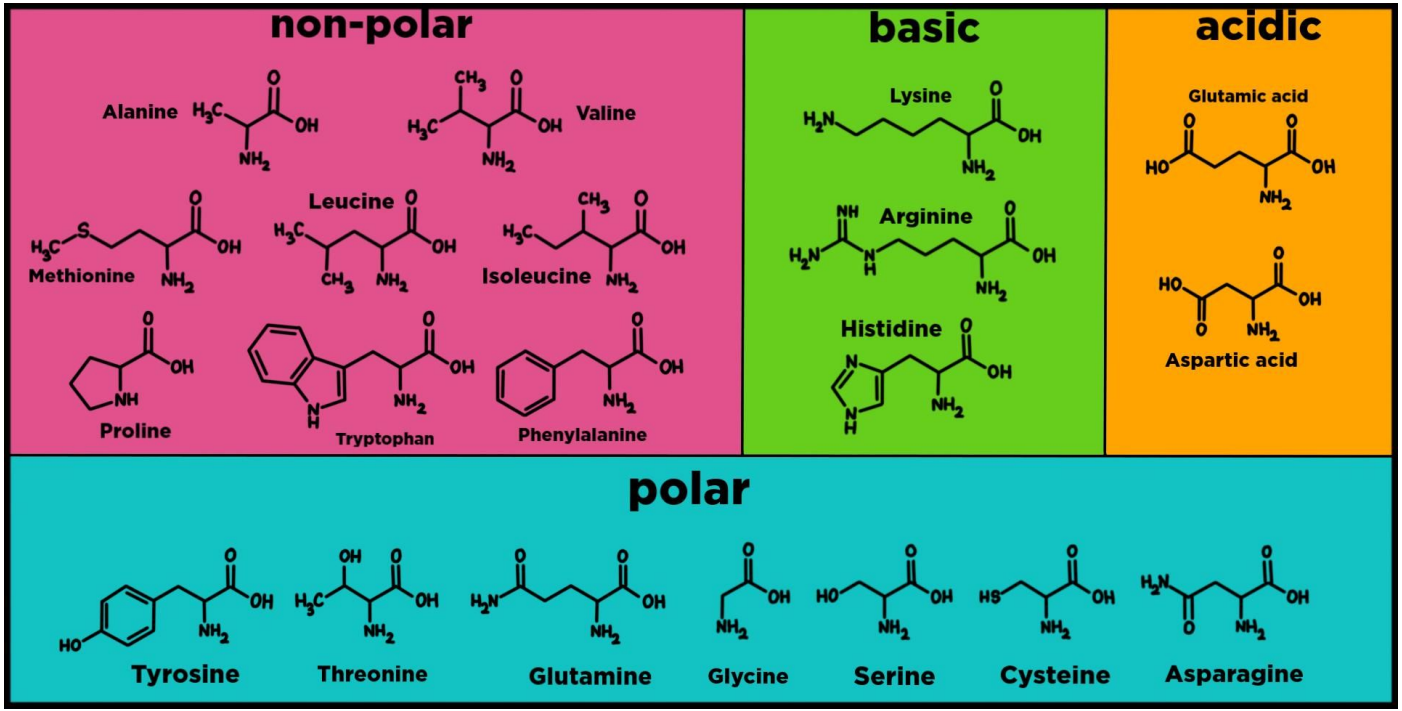
There are a LOT of possible k-mers of DNA

K	Number of possible kmers	Possible k-mers (with DNA)
1	4	A, C, G, T
2	16	AA, AC, AG, AT, CA, CC, CG, CT, GA, GC, GG, GT, TT, TA, TC, TG
3	64	AAA, AAC, AAG, AAT, ACA, ACC, ACG, ACT, AGA, AGC, AGG, AGT, ATA, ATC, ATG, ATT, CAA, CAC, CAG, CAT, CCA, CCC, CCG, CCT, CGA, CGC, CGG, CGT, CTA, CTC, CTG, CTT, GAA, GAC, GAG, GAT, GCA, GCC, GCG, GCT, GGA, GGC, GGG, GGT, GTA, GTC, GTG, GTT, TAA, TAC, TAG, TAT, TCA, TCC, TCG, TCT, TGA, TGC, TGG, TGT, TTA, TTC, TTG, TTT,

There are a LOT of possible k-mers of DNA: 4^k



Even more possible amino acid k-mers: 21^k



Possible k-mers, total k-mers, distinct k-mers, unique k-mers

C A T C G A T C G A

	k = 3
Possible k-mers	
Total k-mers	
Distinct k-mers	
Unique k-mers	

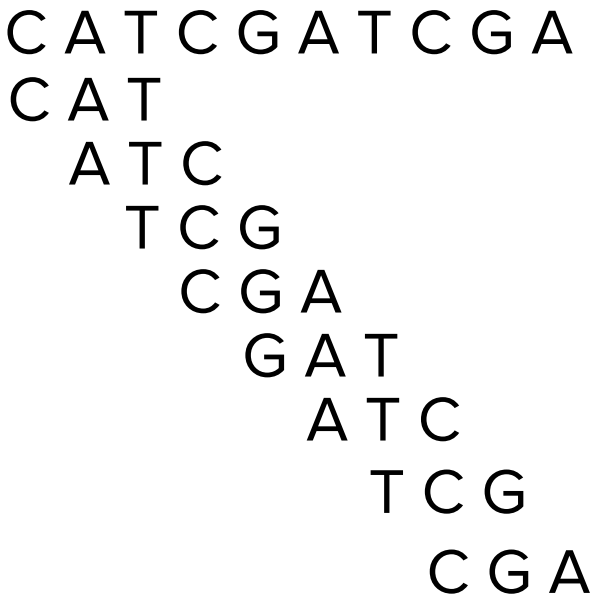
Possible k-mers, total k-mers, distinct k-mers, unique k-mers

	k = 3	
Possible k-mers	4^k	64
Total k-mers		
Distinct k-mers		
Unique k-mers		

C A T C G A T C G A

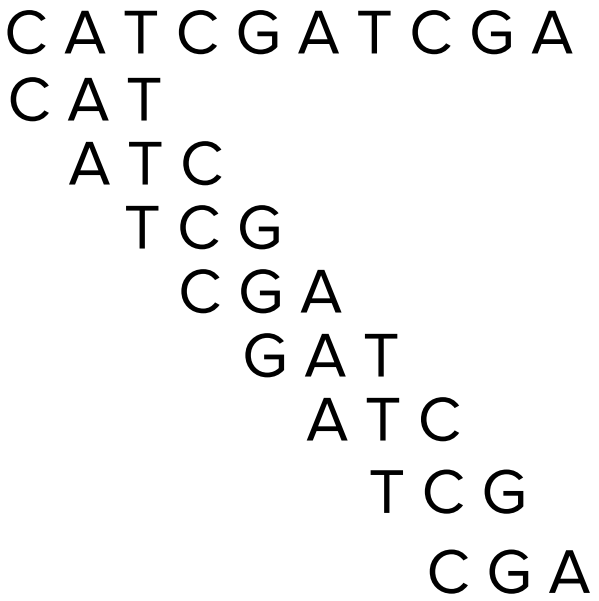
Possible k-mers, total k-mers, distinct k-mers, unique k-mers

	k = 3	
Possible k-mers	4^k	64
Total k-mers		
Distinct k-mers		
Unique k-mers		



Possible k-mers, total k-mers, distinct k-mers, unique k-mers

	k = 3	
Possible k-mers	4^k	64
Total k-mers	l-k+1	10-3+1 = 8
Distinct k-mers		
Unique k-mers		



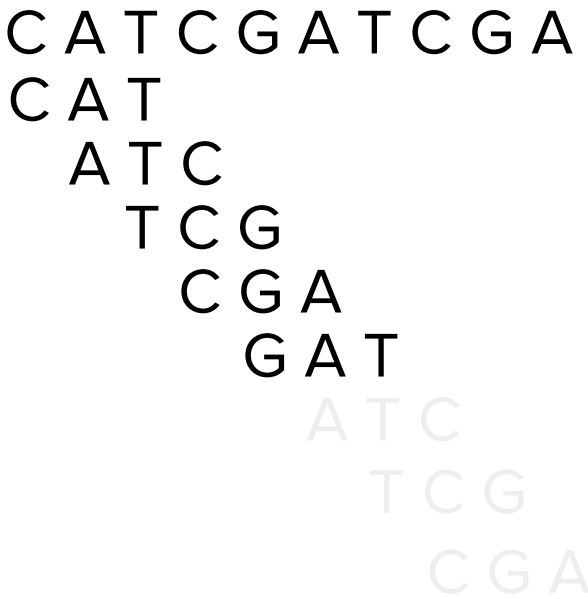
Possible k-mers, total k-mers, distinct k-mers, unique k-mers

	k = 3	
Possible k-mers	4^k	64
Total k-mers	l-k+1	10-3+1 = 8
Distinct k-mers	Counted once, even if they appear multiple times	
Unique k-mers		

```
C A T C G A T C G A
C A T
  A T C
    T C G
      C G A
        G A T
          A T C
            T C G
              C G A
```

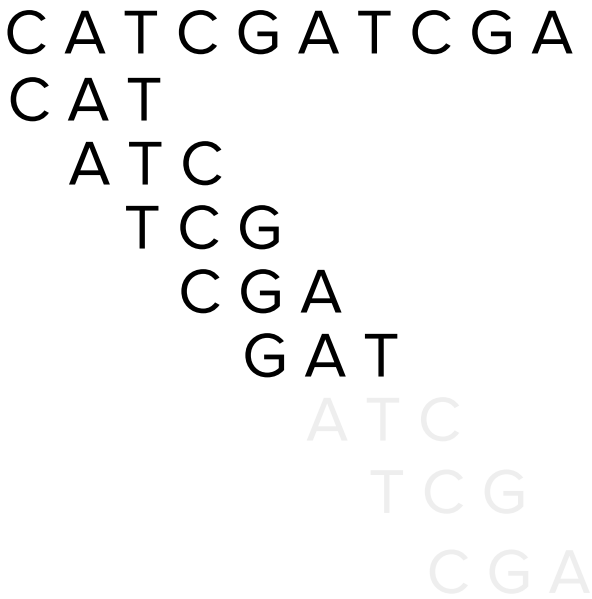
Possible k-mers, total k-mers, distinct k-mers, unique k-mers

	k = 3	
Possible k-mers	4^k	64
Total k-mers	l-k+1	10-3+1 = 8
Distinct k-mers	Counted once, even if they appear multiple times	5
Unique k-mers		



Possible k-mers, total k-mers, distinct k-mers, unique k-mers

	k = 3	
Possible k-mers	4^k	64
Total k-mers	l-k+1	10-3+1 = 8
Distinct k-mers	Counted once, even if they appear multiple times	5
Unique k-mers	Only occur once	



Possible k-mers, total k-mers, distinct k-mers, unique k-mers

	k = 3	
Possible k-mers	4^k	64
Total k-mers	l-k+1	10-3+1 = 8
Distinct k-mers	Counted once, even if they appear multiple times	5
Unique k-mers	Only occur once	2



Expected unique k-mers in the human genome

- The expected number of occurrence of a given k-mer in a monoploid genome of size G is **G/4^k**



Human haploid genome is ~3 billion bp

K	Possible k-mers	Expected #
5	1,024	2,929,687.5
9	262,144	~11,444.09179
13	67,108,864	~44.70348
17	17,179,869,184	~0.174622
21	4,398,046,511,104	~0.00068

How does this relate to scalable genomics and pangenomics?

Sun: Intro to k-mers + Pangenomics

Mon: Genome profiling (Kamil), Fundamentals of kmers and de bruijn graphs (Katie), kmer counting (Kamil)

Tue: Manipulating k-mers (Kamil), Graph kmers (Antoine)

Wed: Building pangenomes (Katie + Vikram), Sketching (Ben Langmead)

Thu: Pangenome case studies (Katie + Vikram), Genome comparisons (Richard + Vikram)

Fri: Mini projects & wrap-up

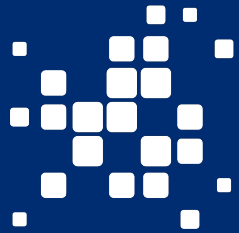


Lex Nederbragt
@lexnederbragt

En réponse à @ctitusbrown

"Finding your way in life is like finding the genome in a De Bruijn graph: it is very easy to find *a* path, very hard to find *the* path".

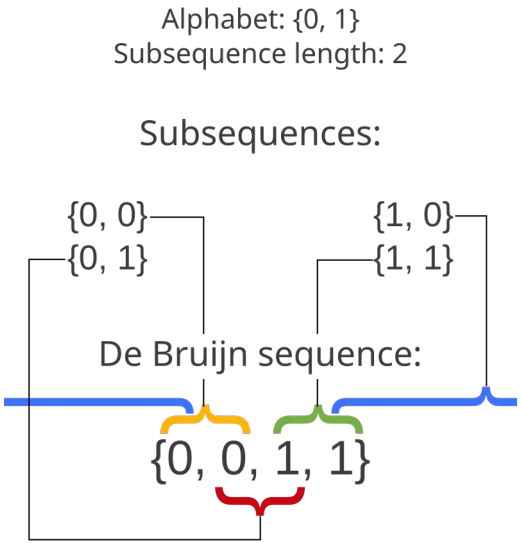
Questions?

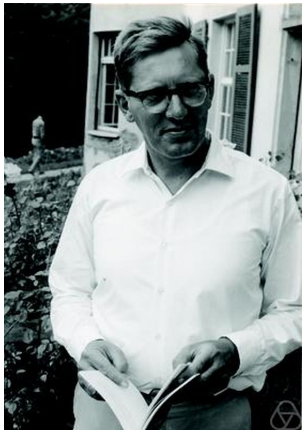


wellcome sanger institute

History of k-mers: de Bruijn sequence

- A sequence where every possible k-tuple appears
- Its length must be: |alphabet|^k
- Not a lot of practical uses



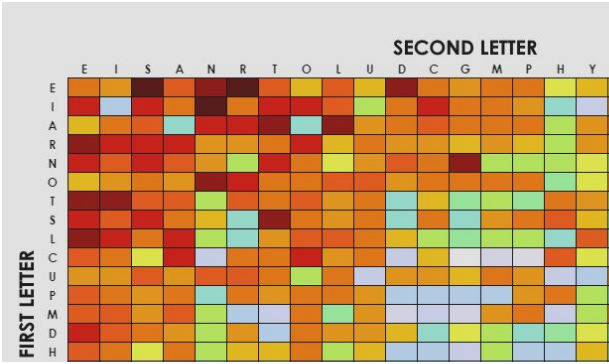


Math interests:

- How to construct it
- Counting how many possible sequences

History of k-mers: Shannon's theory

- Given **frequencies** of English letters
- Generate a "plausible" text, drawing at random:
  - OCRO HLI RGWR NMIELWIS EU LL NBNESEBYA TH EEI …
- Now knowing the **2-mer** frequencies in English texts:
  - ON IE ANTSOUTINYS ARE T INCTORE ST BE S DEAMY …
- The **3-mer** frequencies:
  - IN NO IST LAT WHEY CRATICT FROURE BIRS GROCID …
- Take-home: **k-mer frequencies encode information**
- n consecutive **words,** not letters:
  - n=2: THE HEAD AND IN FRONTAL ATTACK ON AN ENGLISH WRITER THAT …



Reprinted with corrections from *The Bell System Technical Journal*, Vol. 27, pp. 379–423, 623–656, July, October, 1948.

https://www.reddit.com/r/dataisbeautiful/comments/nnxfcr/oc_the_number_of_times_every_twoletter/

A Mathematical Theory of Communication

By C. E. SHANNON

INTRODUCTION

History of k-mers: Or n-grams?



gram means “small weight”

1. Zero-order approximation (symbols independent and equiprobable).

   XFOML RXKHRJFFJUJ ZLPWCFWKCYJ FFJEYVKCQSGHYD QPAAMKBZAACIBZL-HJQD.

2. First-order approximation (symbols independent but with frequencies of English text).

   OCRO HLI RGWR NMIELWIS EU LL NBNESEBYA TH EEI ALHENHTTPA OOBTTVA NAH BRL.

3. Second-order approximation (digram structure as in English).

   ON IE ANTSOUTINYS ARE T INCTORE ST BE S DEAMY ACHIN D ILONASIVE TU-COOWE AT TEASONARE FUSO TIZIN ANDY TOBE SEACE CTISBE.

4. Third-order approximation (trigram structure as in English).

   IN NO IST LAT WHEY CRATICT FROURE BIRS GROCID PONDENOME OF DEMONS-TURES OF THE REPTAGIN IS REGOACTIONA OF CRE.

5. First-order word approximation. Rather than continue with tetragram, . . . , *n*-gram structure it is easier and better to jump at this point to word units. Here words are chosen independently but with their appropriate frequencies.

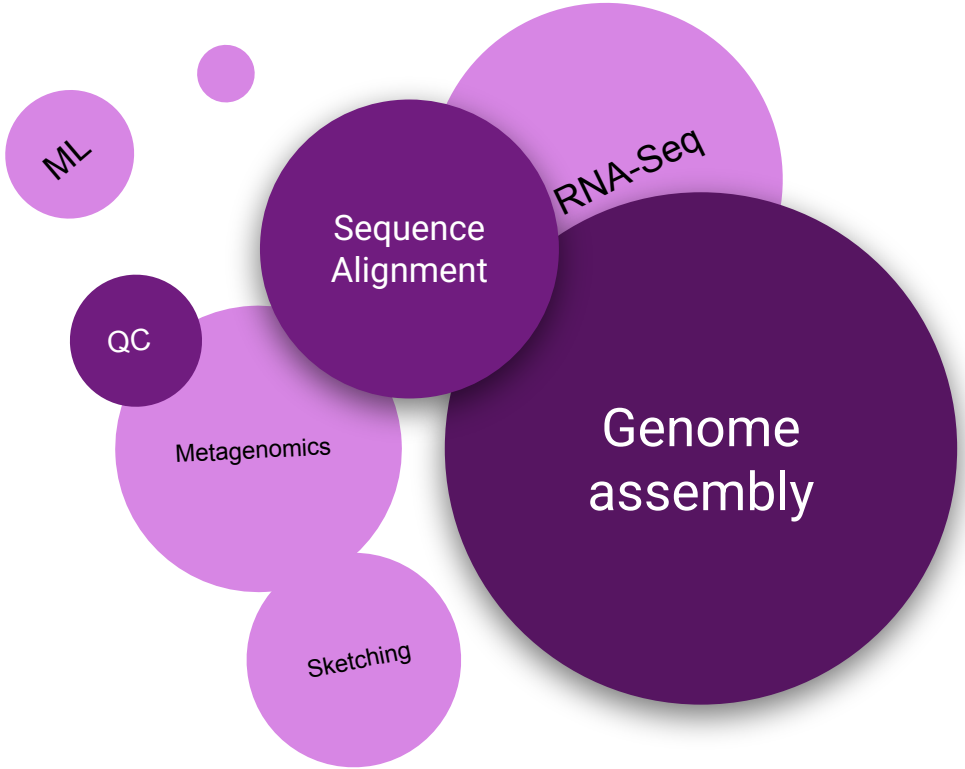
   REPRESENTING AND SPEEDILY IS AN GOOD APT OR COME CAN DIFFERENT NAT-URAL HERE HE THE A IN CAME THE TO OF TO EXPERT GRAY COME TO FURNISHES THE LINE MESSAGE HAD BE THESE.

6. Second-order word approximation. The word transition probabilities are correct but no further struc-ture is included.

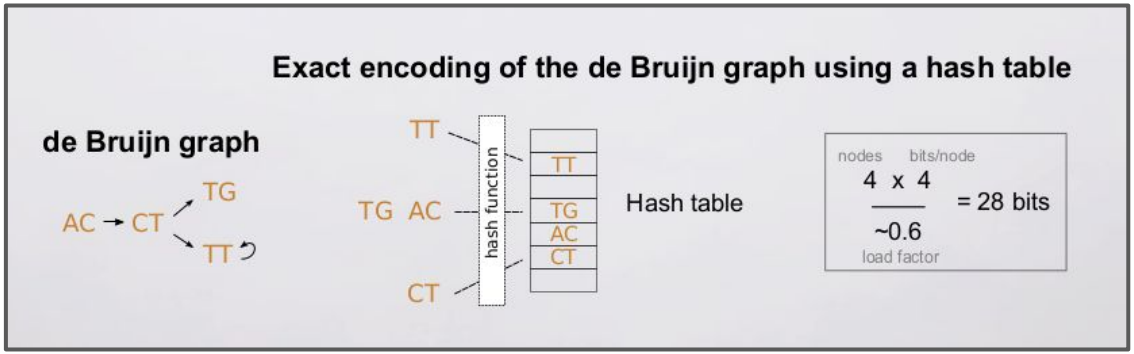
   THE HEAD AND IN FRONTAL ATTACK ON AN ENGLISH WRITER THAT THE CHAR-ACTER OF THIS POINT IS THEREFORE ANOTHER METHOD FOR THE LETTERS THAT THE TIME OF WHO EVER TOLD THE PROBLEM FOR AN UNEXPECTED.

–A Mathematical Theory of Communication, 1948

It’s a k-mer world!



K-mer storage optimization



RESEARCH-ARTICLE OPEN ACCESS

Data Structures to Represent a Set of *k*-long DNA Sequences

Authors: Rayan Chikhi, Jan Holub, Paul Medvedev Authors Info & Affiliations

ACM Computing Surveys, Volume 54, Issue 1 • April 2021 • Article No.: 17, pp 1–22 • https://doi.org/10.1145/3445967

Published: 08 March 2021

The birth of a line of research (2011-2012)

- Conway-Bromage (2011) proposed to encode dBG as bit vector
- Bit vector is of size 4^k, **1**s at positions of *k*-mers
- Efficient succinct encoding (Okanohara *et al* 2006): $O(nk)$
- Info-theoretically optimal



K-mer storage on disk

- No dedicated file format, until recently
- **KFF** (Kmer File Format) (Dufresne et al, 2022)
  - Implemented in KMC, DSK, some other tools

Otherwise:

- .fasta.txt
- HDF5 (not recommended)

